Physics 225, Spring 2026: Homework 6 - Solutions
Due 3/5/26

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
Section:

1. **Cyclic property of the trace.** (30 points) Recall from HW 3 that the trace of a matrix M is the sum of the diagonal entries, which in index notation is $M^i{}_i$. The symbol for the trace of M is $\mathrm{Tr}(M)$.

   (a) Show that $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ using index notation. *Hint: first write out the matrix multiplication AB in index notation, then contract the upper and lower indices to take the trace, and finally rearrange to find the contraction corresponding to BA.*
   Writing the entries of AB as $(AB)^i{}_j = A^i{}_k B^k{}_j$, we have

   $$\mathrm{Tr}(AB) = (AB)^i{}_i = A^i{}_k B^k{}_i = B^k{}_i A^i{}_k = (BA)^k{}_k = \mathrm{Tr}(BA).$$

   The key step is noting that the northeast contractions can be read either as AB (contracting k) or BA (contracting i).

   (b) Show that $\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA) = \mathrm{Tr}(CAB)$. *Hint: use the result from part (a) and the associativity of matrix multiplication to group the product into 2 terms.*
   Since $ABC = A(BC)$, we can let $BC = M$ (since it's just another matrix) and use the result from part (a):

   $$\mathrm{Tr}(ABC) = \mathrm{Tr}(AM) = \mathrm{Tr}(MA) = \mathrm{Tr}(BCA).$$

   We can then apply exactly the same steps with $CA = N$ to $\mathrm{Tr}(BCA)$:

   $$\mathrm{Tr}(BCA) = \mathrm{Tr}(BN) = \mathrm{Tr}(NB) = \mathrm{Tr}(CAB).$$

   (c) Show that the result from part (b) generalizes to an arbitrary number of terms in the product: $\mathrm{Tr}(A_1 A_2 \cdots A_{n-1} A_n) = \mathrm{Tr}(A_n A_1 A_2 \cdots A_{n-1})$. This is called the *cyclic property of the trace* because the trace remains the same after any matrix is cycled from the end of a product of matrices to the beginning.
   Let $M = A_1 \cdots A_{n-1}$ and use $\mathrm{Tr}(MA_n) = \mathrm{Tr}(A_n M)$, and the desired result follows.

2. **The orthogonal group.** (40 points) The set of $n \times n$ orthogonal matrices M can be *defined* as those satisfying $M^{\mathrm{T}} M = I$, where I is the $n \times n$ identity matrix. Show that this set of matrices forms a group for any n, using *only* the defining relationship $M^{\mathrm{T}} M = I$ and the rules for matrix multiplication with transposes and inverses as discussed in lecture. This group is called the *orthogonal group* $\mathrm{O}(n)$. *Hint: the goal is to show that the identity element, the inverse element, and the product of two group elements all satisfy the defining relationship,*

thus establishing that they belong to the group. In these calculations, you will probably find that matrix notation, rather than index notation, is more convenient.

First, we will show that the identity matrix I belongs to the group: $I^{\mathrm{T}}I = II = I$, so I satisfies the defining relationship. Next, if M is an orthogonal matrix, its inverse is $M^{-1} = M^{\mathrm{T}}$, and $(M^{\mathrm{T}})^{\mathrm{T}}M^{\mathrm{T}} = MM^{\mathrm{T}} = I$ for an orthogonal matrix as shown in lecture, so M^{-1} satisfies the defining relationship. Finally, if M and N are orthogonal matrices, then $(MN)^{\mathrm{T}}(MN) = N^{\mathrm{T}}M^{\mathrm{T}}MN$. Since $M^{\mathrm{T}}M = I$ and $N^{\mathrm{T}}N = I$ from the definition of M and N as orthogonal matrices, we have $N^{\mathrm{T}}(M^{\mathrm{T}}M)N = N^{\mathrm{T}}IN = N^{\mathrm{T}}N = I$. Thus $(MN)^{\mathrm{T}}(MN) = I$, and a product of orthogonal matrices is also an orthogonal matrix, establishing the closure property.

3. **Affine transformations form a group.** (30 points) Show that the affine transformations from HW 5, problem 3 form a group. *Hint: be careful about how you define the inverse element! The ordered-pair $(R(\theta), \vec{a})$ representation is easiest for this purpose, rather than the 3×3 matrix representation.*

In the ordered-pair representation, $A = (R(\theta), \vec{a})$ acts on a 2-component vector $\vec{v}$ as $\vec{v} \xrightarrow{A} R(\theta)\vec{v} + \vec{a}$. The identity element is $I = (I_2, \vec{0})$, since applying this after A gives

$$\vec{v} \xrightarrow{IA} I_2(R(\theta)\vec{v} + \vec{a}) + \vec{0} = I_2R(\theta)\vec{v} + I_2\vec{a} + \vec{0} = R(\theta)\vec{v} + \vec{a}$$

which is the same as just applying A. This also implies that I acting on $\vec{v}$ just gives $\vec{v}$. The inverse element A^{-1} is such that $\vec{v} \xrightarrow{A^{-1}A} \vec{v}$. It's tempting to think this is just the inverse of the two components of A, namely $(R(-\theta), -\vec{a})$, but this is not quite right:

$$R(-\theta)(R(\theta)\vec{v} + \vec{a}) - \vec{a} = \vec{v} + R(-\theta)\vec{a} - a \neq \vec{v}$$

We can fix the mismatch by choosing $A^{-1} = (R(-\theta), -R(-\theta)\vec{a})$ instead:

$$\vec{v} \xrightarrow{A^{-1}A} = R(-\theta)(R(\theta)\vec{v} + \vec{a}) - R(-\theta)\vec{a} = \vec{v} + R(-\theta)\vec{a} - R(-\theta)\vec{a} = \vec{v}.$$

The closure property is easily established by taking $B = (R(\phi), \vec{b})$ and considering

$$\vec{v} \xrightarrow{AB} = R(\phi)(R(\theta)\vec{v} + \vec{a}) + \vec{b} = R(\phi)R(\theta)\vec{v} + (R(\phi)\vec{a} + \vec{b}).$$

By your work in Discussion 6, we know that $R(\phi)R(\theta) = R(\phi + \theta)$, and $R(\phi)\vec{a} + \vec{b}$ is just another 2-component vector which we could call $\vec{c}$. Thus AB can be represented by an ordered pair $(R(\phi + \theta), \vec{c})$, which has the correct form of an affine transformation.